

11 One-Loop Radiative Corrections in Quantum Electrodynamics

Problems

1. Calculate the contributions to the vacuum polarization function $\pi(q^2)$ and to Z_3 of one-loop graphs involving a charged spinless particle of mass m_s . What effect does this have on the energy shift of the $2s$ state of hydrogen, if $m_s \gg Z\alpha m_e$?
2. Suppose that a neutral scalar field ϕ of mass m_ϕ has an interaction $g\phi\bar{\psi}\psi$ with the electron field. To one-loop order, what effect does this have on the magnetic moment of the electron? On Z_2 ?
3. Consider a neutral scalar field ϕ with mass m_ϕ and self-interaction $g\phi^3/6$. To one-loop order, calculate the S -matrix element for scalar-scalar scattering.
4. To one-loop order, calculate the effect of the neutral scalar field of Problem 2 on the mass shift δm_e of the electron.

References

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